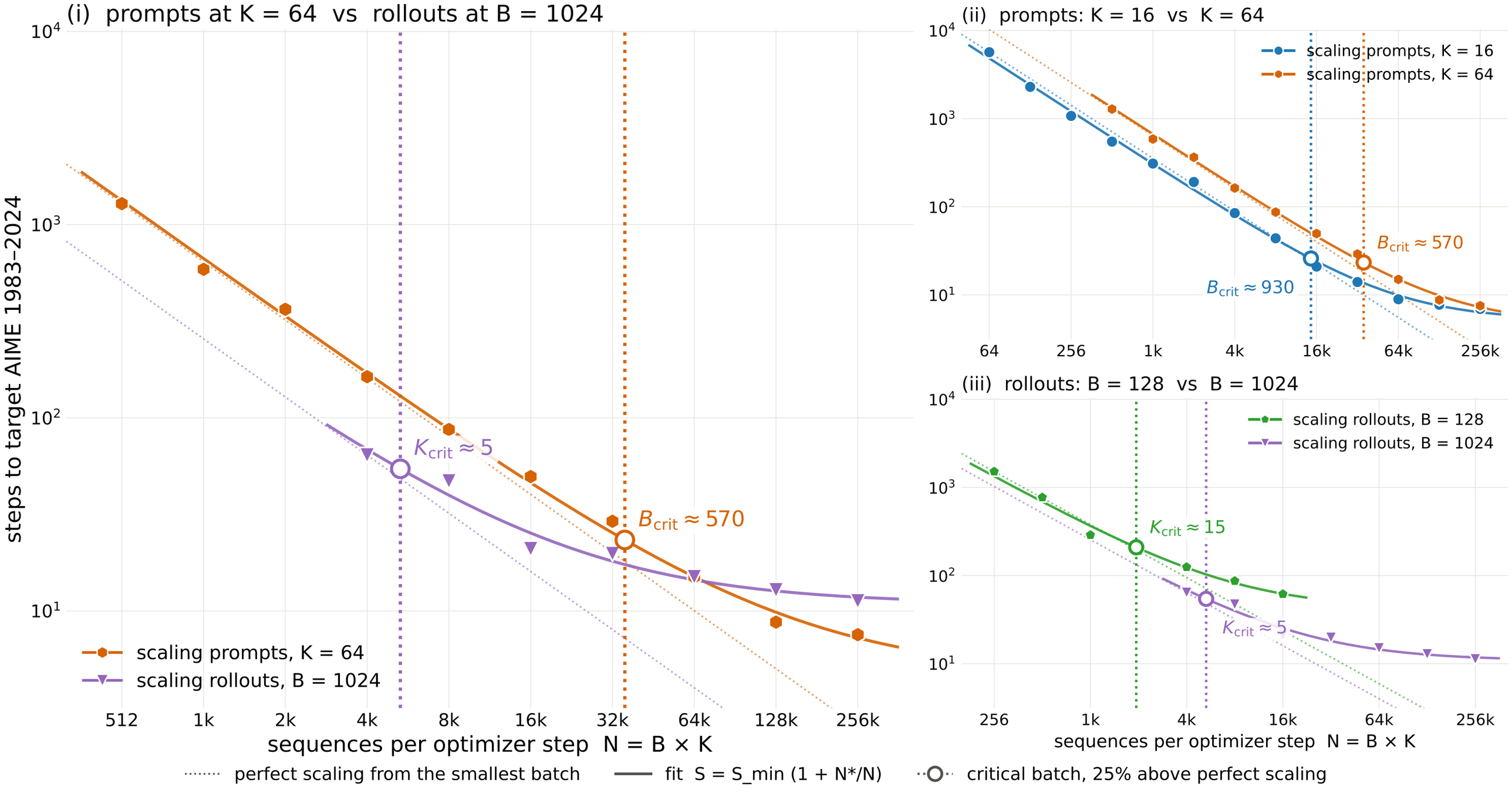
(i) prompts at K = 64 vs rollouts at B = 1024
steps to target AIME 1983–2024
10^4
10^3
10^2
10^1
$K_{crit} \approx 5$
$B_{crit} \approx 570$
scaling prompts, K = 64
scaling rollouts, B = 1024
512
1k
2k
4k
8k
16k
32k
64k
128k
256k
sequences per optimizer step N = B × K
(ii) prompts: K = 16 vs K = 64
scaling prompts, K = 16
scaling prompts, K = 64
$B_{crit} \approx 930$
$B_{crit} \approx 570$
64
256
1k
4k
16k
64k
256k
(iii) rollouts: B = 128 vs B = 1024
scaling rollouts, B = 128
scaling rollouts, B = 1024
$K_{crit} \approx 15$
$K_{crit} \approx 5$
256
1k
4k
16k
64k
256k
sequences per optimizer step N = B × K
perfect scaling from the smallest batch
fit S = S_min (1 + N*/N)
critical batch, 25% above perfect scaling